

Officiating, Replay, and the Rules of the Ball

Seven officials, three ways into the replay booth, and the handful of rules that decide who has the ball, so you can call it before the referee turns on his microphone.

Fourth quarter, 1:41 left in a one-score game. A receiver dives along the sideline, the ball pinned to his chest, and tumbles out of bounds. The nearest official, a man in a black cap twenty yards downfield, waves both arms across his body: incomplete. The home crowd roars at the stadium screen, which is replaying the catch from a low angle. The losing coach grabs the red flag from his sock, then stops; he isn't allowed to throw it. Thirty seconds later, with nobody on either sideline having asked, the official in the white cap walks to the middle of the field, clicks on his microphone and says: *"After review, the ruling on the field is reversed. The receiver had control of the ball with both feet down in bounds. The pass is complete."* Who decided that? Why couldn't the coach challenge? And what exactly did "control with both feet down" have to do with it? This chapter is about the people who run the game and the small set of rules that decide, on almost every close play, who has the ball.

You'll be able to...

- Name the seven officials, find each one on a wide shot, and say which fouls he is most likely to throw a flag for.
- Say, for any close play, whether it can be reviewed and by whom: a coach's challenge, a booth review, replay assist, or nobody.
- Rule on a catch the way the replay official will (control, feet, then a football act), and say why the league rewrote the rule in 2018.
- Apply the rules of the ball: down by contact, forward progress, fumble versus incomplete pass, the fumble through the end zone, and backward versus forward passes.
- Call intentional grounding before the flag lands, using the tackle box and the line of scrimmage.
- Decode a referee's announcement and a virtual measurement, and predict "stands", "confirmed" or "reversed" before he says it.

Seven officials, seven jobs

An NFL game is run by seven officials on the field, each wearing a number and a letter on his back that names his job. A seven-person crew sounds like a lot until you count what happens on one snap: 22 players, most of them in contact with someone, spread across 53 yards of width and, on a deep pass, 50 yards of depth, all in about five seconds. Nobody can watch that. So the crew splits the field, and the players, into pieces. Each official has a starting spot, a few players he is responsible for (his *keys*), an area he must cover once the ball moves, and a short list of fouls that are his to call. The aim is that every player, every inch of sideline and every goal line has at least one official looking at it, and the most important ones have two.

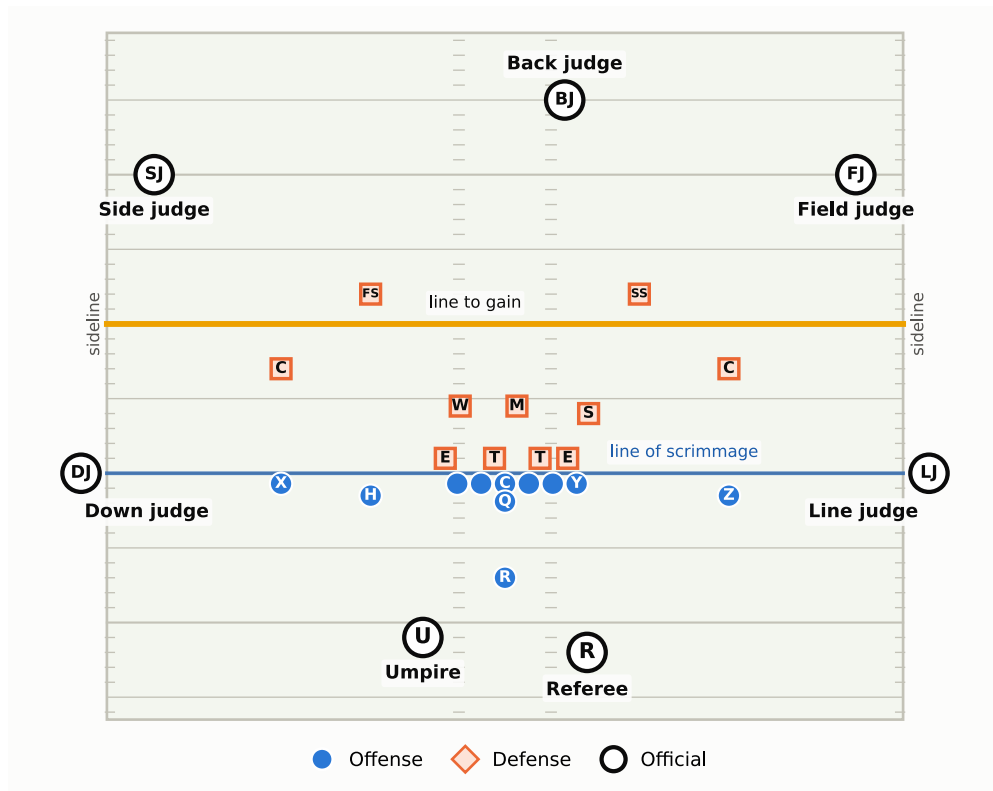


Figure 1: Where the seven officials stand before a snap (white discs; an 11-personnel offense in blue against a 4-3 defense in orange). Two officials work behind the offense: the referee (R), on the side of a right-handed quarterback's throwing arm, and the umpire (U) opposite him. Two straddle the line of scrimmage at the sidelines, just off the field: the down judge (DJ), on the side with the chain crew, and the line judge (LJ). Three work deep: the side judge (SJ) and field judge (FJ) about 20 yards downfield near each sideline, and the back judge (BJ) about 25 yards deep in the middle. Exact depths vary with the down, the distance and the formation.

Here is what each of them is watching. The table is the short version of the league's own description of the positions;¹ the habits behind it are what let you guess who threw a flag before the referee says.

¹The positions and duties follow the NFL's own descriptions ("Officials' responsibilities and positions," NFL Football Operations), as reproduced in Crescent City Sports, "Breakdown of duties for NFL game officials" (Oct. 20, 2019): referee and umpire 10–12 yards deep in the backfield, side judge and field judge 20 yards downfield, back judge 25 yards: <https://crescentcitysports.com/breakdown-of-duties-for-nfl-game-officials/>; "Official (American football)," Wikipedia, for the referee favoring the quarterback's throwing arm and the seven abbreviations: [https://en.wikipedia.org/wiki/Official_\(American_football\)](https://en.wikipedia.org/wiki/Official_(American_football)).

Table 1: The seven officials: where they start, what they watch, and the calls you'll most often see them make.

Official	Starts	Watches	Typical calls
Referee (R, white cap)	Offensive backfield, 10–12 yards deep, on the quarterback's throwing-arm side	The quarterback and the backs; the snap	Roughing the passer, intentional grounding, fumble or pass by the quarterback; announces every foul and review
Umpire (U)	Offensive backfield, opposite the referee; on the defensive side for field goals and extra points	The center, guards and the men across from them	Holding (both sides) and false starts inside, illegal hands to the face; spots the ball; checks equipment
Down judge (DJ)	On the line of scrimmage at one sideline, with the chain crew	The line (who's on it, who moved), the end and receivers on his side, his sideline	Offside, encroachment, false start, illegal formation or motion; forward progress and out of bounds on his side
Line judge (LJ)	On the line of scrimmage at the other sideline	The same as the down judge, on his side	The same, plus whether a pass was thrown from beyond the line
Side judge (SJ)	About 20 yards deep near the down judge's sideline	Receivers on his side once they're downfield	Pass interference, illegal contact, catch or no catch, out of bounds
Field judge (FJ)	About 20 yards deep near the line judge's sideline	Receivers on his side once they're downfield	The same; under one upright on kicks
Back judge (BJ)	About 25 yards deep in the middle	The tight end and inside receivers, the deep middle, the end line	Deep pass interference, catches in the end zone; under the other upright on kicks

A few things in that table explain a lot of what you see on Sundays.

The wing officials own the line of scrimmage. The **down judge** and **line judge** stand exactly on the line at the sidelines so they can look straight down it. That is the only place from which you can see who is offside, which receivers are on the line and which are off it (the formation rules need seven on the line), and whether a quarterback crossed the line before he threw. Most pre-snap fouls (the false starts and offsides) come from these two. After the snap they drift downfield along the sideline and mark where a runner's **forward progress** stopped: the spot you see an official jog in and point his foot at.

The umpire lives in the trenches, and he moved. For most of football history the **umpire**, the official responsible for the interior line, stood four or five yards behind the defensive line, in the middle of the linebackers, which is where the holding happens. It was also where umpires kept getting run over. In 2010 the NFL moved him to the offensive backfield, opposite the referee, where he watches the same linemen from behind.² The trade was visibility for safety: from behind you see a lineman's hands on a jersey less clearly. And it created a new problem, because the umpire also spots the ball after most plays, and walking from the offensive backfield to the ball slows the next snap. For that reason, from 2010 to 2015 he went back to his old spot behind the defense in the last five minutes of the game, when an offense wants to snap fast. Since 2016 he has stayed in the backfield on every snap from scrimmage; only on field goals and extra points (since 2023) does he work on the defensive side.

The deep officials are pass officials. The **side judge**, **field judge** and **back judge** are the reason a deep throw can be judged at all. Each takes a share of the receivers, keeps them between himself and the sideline or the end line, and rules on pass interference and illegal contact, catches and out of bounds. The seventh official, the side judge, was added in 1978, the same year the league

²The umpire was moved to the offensive backfield for the 2010 season after a run of injuries to umpires in the middle of the defense (five serious ones in 2009). From 2010 to 2015 he returned to the defensive side in the last five minutes of the game; that exception was dropped in 2016, and since 2023 he works on the defensive side on field goals and extra points. "Official (American football)," Wikipedia (umpire): [https://en.wikipedia.org/wiki/Official_\(American_football\)](https://en.wikipedia.org/wiki/Official_(American_football)).

loosened the rules on pass defense to open up the passing game: more passing meant more eyes downfield.³

The referee is the boss, and the voice. The **referee** wears the white cap and leads the crew. His spot, behind and to the throwing-arm side of the quarterback, lets him watch the one player the rules protect most: he is the official who throws most roughing-the-passer flags, and the one who decides whether a quarterback who lost the ball was throwing it or fumbling it. When a flag lands, any official can have thrown it, but the crew gathers, the official who saw the foul tells the referee what it was, and the referee announces it. A flag thrown by one official and waved off after the conference (“picked up”) usually means another official had a better look.

i Madden vs. real life

In a video game a foul is a fact: the code knows a lineman held, and a flag appears. On a real field, a foul exists only if an official in the right spot sees it. A hold on the far side of a run play can be in no official’s view, and a hold on the play side might be seen by two. That is why “they’re letting them play” is sometimes true, why the same contact draws a flag in front of the line judge and not behind the back of the umpire, and why teams study which way each crew’s eyes point. Officiating is a sample, not a census.

The crew in motion

Starting spots are only half of it. Watch a wide shot of a deep pass and you’ll see the crew move as a unit: the referee stays with the quarterback, the wing officials hold the line until the ball is thrown (to see whether it was thrown from beyond it) and then run downfield, and the deep officials backpedal to stay deeper and wider than the receivers they’re judging, so that every catch is seen from the side or from in front, never from behind.

³The NFL went to seven officials in 1978 by adding the side judge; the back judge and field judge exchanged titles in 1998 to match college and high-school usage; and the head linesman was renamed the down judge in 2017. “Official (American football),” Wikipedia (history): [https://en.wikipedia.org/wiki/Official_\(American_football\)](https://en.wikipedia.org/wiki/Official_(American_football)) ; “1998 NFL season,” Wikipedia: https://en.wikipedia.org/wiki/1998_NFL_season. The 1978 changes to pass-defense rules are covered in The 1978 Liberation.

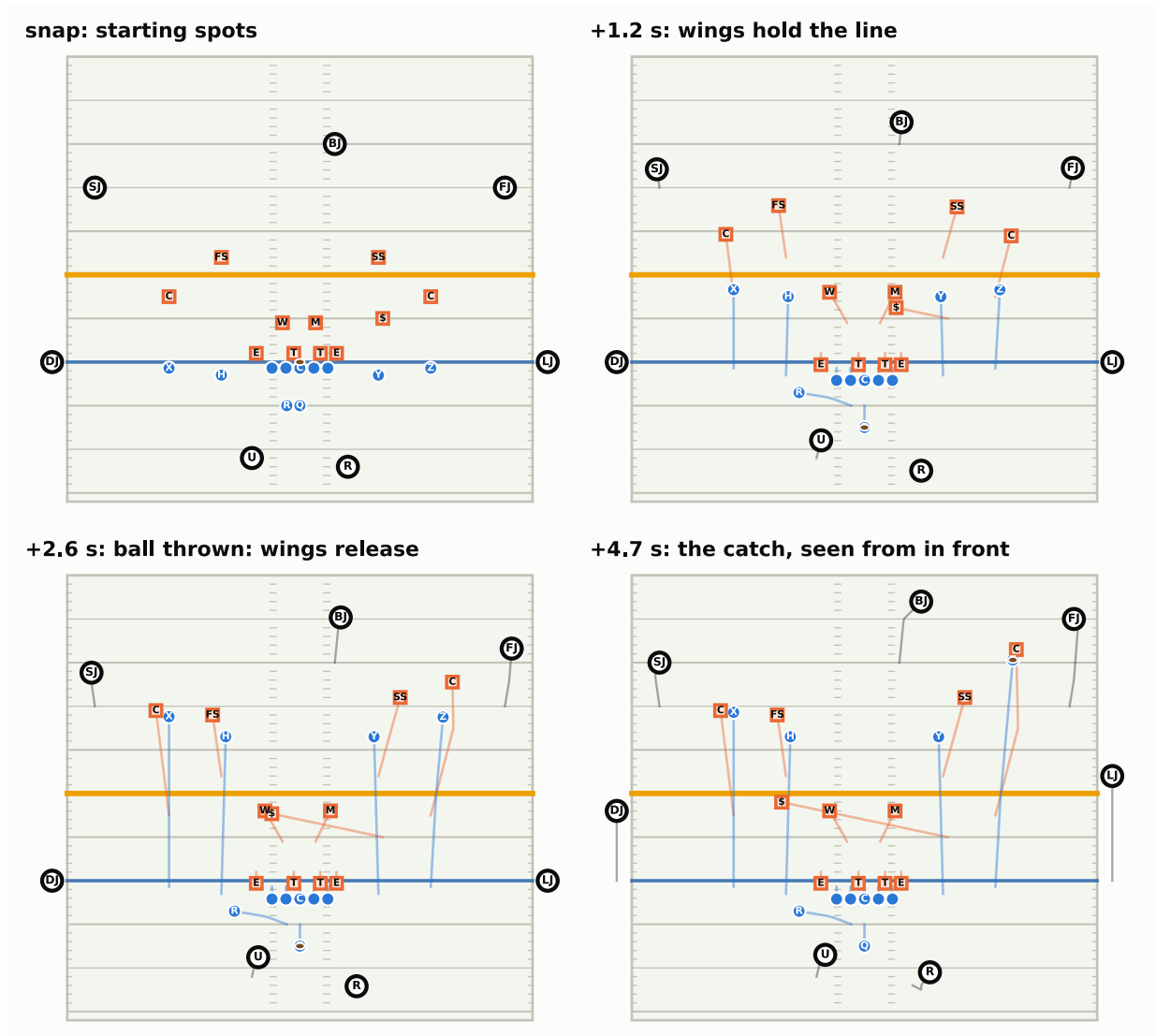


Figure 2: A deep pass seen through the officials (simulated tracking in Big Data Bowl format; an illustrative play, not a real one). Snap: the crew in its starting spots. 1.2 s: the quarterback is in his drop and the receivers are climbing; the referee (R) watches the quarterback, the umpire (U) the linemen, and the down judge (DJ) and line judge (LJ) still hold the line, because only from there can they judge whether the pass was thrown from behind it. 2.6 s: the ball is thrown to the receiver on the right sideline; the wing officials release downfield and the field judge (FJ) backpedals to stay wider and deeper than the receiver. 4.7 s: the catch, judged by the field judge from in front and outside, with the line judge trailing to mark the spot. Faint lines are each person's path.

Where the voice came from

The referee's microphone is younger than you might think. For most of the game's history fans learned what a flag was for from the referee's arm signals and the next morning's newspaper. In 1975, on the watch of the league's supervisor of officials, Art McNally, referees were fitted with wireless microphones so they could tell the stadium, and the television audience, what had been called. McNally, a Philadelphian who had officiated thousands of games in several sports before he ran NFL officiating from 1968 to 1991, spent his career trying to make officiating explainable: he also oversaw the first instant-replay system in 1986, and the league's officiating command center in New York is named after him. In 2022 he became the first official elected to the Pro Football Hall of Fame.⁴

⁴Pro Football Hall of Fame, Art McNally (Class of 2022, the first game official enshrined; NFL supervisor of officials 1968–1991; the “Father of Instant Replay”): <https://www.profootballhof.com/players/art-mcnally/> ; Pro Football Hall of Fame, “1975: Referees are equipped with wireless microphones”: <https://www.profootballhof.com/football-history/football-history/1960-1979/1975> ; Football Zebras, “Referee announcements have come a long way in 38 years” (Oct. 2013): <https://www.footballzebras.com/2013/10/referee-announcements-have-come-a-long-way-in-38-years/>. The league's New York officiating center is “Art McNally GameDay Central”: <https://operations.nfl.com/rules-officiating/instant-replay/art-mcnally-gameday-central>.

Replay: three doors into the booth

Officials miss things; they are watching a five-second play at full speed from one angle. Replay exists to fix the misses that matter most without stopping the game every few minutes, and the NFL's answer is a system with three separate doors, each with its own rules about who may open it and what it can change.

First, the history, because each door was added to fix a problem the last one created.⁵

- **1986–1991: the first try.** An official in the press box could stop the game to review a play. There were no challenges. The system was slow, overturned little, and was voted out after 1991.
- **1999: the challenge flag.** Replay came back with a twist: the *coaches* would decide which plays to review, by throwing a red flag, and pay for a failed challenge with a timeout. The booth kept control of the final two minutes of each half, so coaches couldn't use challenges as free timeouts when the clock matters most.
- **2011 and 2012: automatic reviews.** Every scoring play (2011) and every turnover (2012) became a booth review, so a coach no longer had to spend a challenge to fix the biggest calls of the game.
- **2017: New York decides.** The final say moved from the referee under a sideline hood to the league's senior officiating staff in New York, with the referee consulting on a tablet, so that the same handful of people judge every review in every game.
- **2019: the pass-interference experiment.** After a missed interference call decided the NFC Championship Game in January 2019 (the story is in Contact Rules and Penalties), pass interference was reviewable for one season. Almost nothing was overturned, and the experiment was dropped.
- **2021: replay assist.** The booth began fixing clear-cut objective facts on its own, without a formal stop, and its powers grew in 2024, 2025 and 2026 (below).

Every review uses the same standard: the ruling on the field is changed only on **clear and obvious** visual evidence. That produces the three outcomes in every announcement. *Reversed* means the video shows the call was wrong. *Confirmed* means the video shows it was right. *Stands* means the video doesn't show enough either way, so the call on the field survives. The distinction matters for your predictions: on a play the camera can't see well (a ball at the bottom of a pile, a knee hidden by a body), the answer is almost always "stands", whatever the call was.

Door 1: the coach's challenge

A **coach's challenge** is the head coach asking for a review by throwing a red flag onto the field before the next snap. The rules make it scarce on purpose.⁶

- Each team gets **two challenges** a game. Since 2024 it gets **a third if either of the first two succeeds**; before that, both had to succeed.
- A team must have a **timeout** left to challenge, and it **loses that timeout if the challenge fails**.
- A coach may not challenge in the **final two minutes of either half** or in **overtime**, or challenge a play the booth reviews automatically (a scoring play or a turnover); those belong to the booth.
- Only certain rulings are reviewable: whether a pass was caught, whether a runner was down, in bounds or across the goal line, the spot relative to the line to gain or the goal line, who recovered a loose ball, too many men on the field. Judgment fouls, holding or interference, are not.

Why make it expensive? Because a free challenge would be thrown on every close play, and each review costs two or three minutes of everyone's time. The timeout charge turns a challenge into a bet: throw the flag when the expected gain (the chance of winning the review times what the reversal is worth) beats the expected cost (the chance of losing times what a timeout is worth later). That is exactly the kind of decision the game-management staff upstairs exists to make, often in the eight or ten seconds before the next snap, with a video coordinator watching the television replay and calling the head coach's headset. It's also why an offense that likes the call on the field hurries to the line: once the ball is snapped, the play can't be challenged (In-Game Adjustments and Tempo).

⁵"Replay review in gridiron football," Wikipedia (NFL section): the 1986–1991 system, the 1999 challenge system, automatic review of scoring plays (2011) and turnovers (2012), centralized decisions in New York (2017), and the one-season pass-interference review (2019): https://en.wikipedia.org/wiki/Replay_review_in_gridiron_football.

⁶NFL Rulebook (2026), Rule 15, Section 1: two challenges, each needing an available timeout; a timeout charged for an unsuccessful challenge; a third challenge if at least one succeeds; a team that challenges when it isn't permitted to is charged a timeout, or 15 yards if it has no timeouts left; only the Replay Official may start a review after the two-minute warning, in overtime, on scores, tries, turnovers and a ruling of short on fourth down: <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>. The 2024 change (a Detroit proposal; before it, both of the first two challenges had to succeed): Pride of Detroit, "Detroit Lions rule proposal on challenges barely accepted" (Mar. 25, 2024): <https://www.prideofdetroit.com/2024/3/25/24111469/detroit-lions-rule-proposal-challenges-barely-accepted-2024-nfl-season>.

And it's why challenges are rarer than you'd guess. In 2025 coaches threw 151 challenges in 272 regular-season games, about 0.28 per team per game, and won 52% of them; over 2016–2025 the success rate was 46%.⁷ A coach who is right about half the time, and is only betting on plays he's fairly sure of, is telling you how hard it is to see a play clearly from the sideline in ten seconds.

Door 2: booth review

A **booth review** is started by the replay official in the stadium, working with the league's officiating command center, not by a coach. It covers:

- **every scoring play** (touchdowns, field goals, safeties) and **every try**;
- **every turnover** (interceptions, fumbles recovered by the defense, and muffed kicks recovered by the kicking team), and every ruling that the offense came up **short on fourth down**;
- **every reviewable play in the final two minutes of each half and in overtime**.

Most booth reviews never stop the game. A touchdown is looked at during the celebration and the try is lined up; if the video agrees, the game goes on. If the booth wants a closer look, it signals the referee (the officials wear a buzzer for this) and the game stops. The coach's flag is irrelevant here: the booth decides whether to look. A **replay review** of either kind ends the same way, with the referee walking to the middle of the field, turning on his microphone, and reading New York's decision.

Door 3: replay assist

The newest door is the quietest. **Replay assist** lets the booth tell the on-field crew about a clear-cut objective fact while the officials are still sorting out the play, with no formal review and usually no stoppage you'd notice: the ball was actually spotted a yard short, the runner's knee was down before the ball came out, the receiver's second foot landed out of bounds, two seconds should go back on the clock. It began in 2021 as a way to fix the obvious without a coach having to gamble a timeout on it.

In 2024 the league gave it its first power over fouls: it could wipe out a roughing-the-passer flag when the video showed no contact with the quarterback's head or neck, a grounding flag when the passer was clearly outside the pocket, and a flag for a hit out of bounds when the runner was clearly still in bounds. In 2025 it added five more. Replay assist may now also **overturn** a flag for a hit on a defenseless player, a face mask, a horse-collar tackle, tripping, or roughing or running into the kicker, when the video plainly shows the foul didn't happen (the hand never touched the face mask, the kicker was never touched). It **cannot add** a foul that nobody called. In 2026 the league added one more: its officiating staff may **consult** with the on-field crew on ejections for flagrant football acts and non-football acts, such as a punch thrown away from the play. Some reports describe that as replay ejecting players on its own; the league's official wording is only that it may consult.⁸

What the NFL still doesn't have is a **sky judge**: an official in the booth with the power to call or wipe out any foul, in real time, the way a video referee works in some other sports. Other leagues have tried versions of it, and the NFL has discussed it for years. The objections are practical: it would slow the game, it would turn every play (on which, coaches like to say, somebody is holding) into a candidate for a flag, and it would put judgment calls in the hands of someone watching a screen rather than the field. So the league has expanded replay one narrow, objective slice at a time.

⁷Course calculation from nflverse play-by-play, 2016–2025: plays flagged `replay_or_challenge`, classified by their description ("[team] challenged the ... ruling" for a coach's challenge, "The Replay Official reviewed the ... ruling" for the booth) and by the result ("REVERSED" or "Upheld"; nflverse text does not separate "stands" from "confirmed"). Challenge figures are regular season only. Reviews that ended quickly without a stoppage are generally not written into the play text.

⁸2021 start: ESPN, "NFL replay reviews, challenges drop with new rules and tech": https://www.espn.com/nfl/story/_/id/35468076/nfl-replay-reviews-challenges-drop-new-rules-tech. 2024 (flags for roughing the passer, grounding, hits out of bounds): CBS Sports, "NFL to expand replay reviews to include intentional grounding, roughing the passer, hit out of bounds" (Mar. 30, 2024): <https://www.cbssports.com/nfl/news/nfl-to-expand-replay-reviews-to-include-intentional-grounding-roughing-the-passer-hit-out-of-bounds>. 2025 (hits on a defenseless player, face mask, horse-collar, tripping, roughing or running into the kicker; no fouls added): CBS Sports, "Five new NFL rules to know heading into Week 1 of preseason": <https://www.cbssports.com/nfl/news/five-new-nfl-rules-to-know-heading-into-week-1-of-preseason-here-are-the-biggest-changes-for-2025>. The current list is in NFL Rulebook (2026), Rule 15, Section 9: <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>. 2026: Seattle Seahawks, "NFL announces approved rule changes for 2026 season" (league personnel may consult on ejections for flagrant football and non-football acts): <https://www.seahawks.com/news/nfl-announces-approved-rule-changes-for-2026-season>.

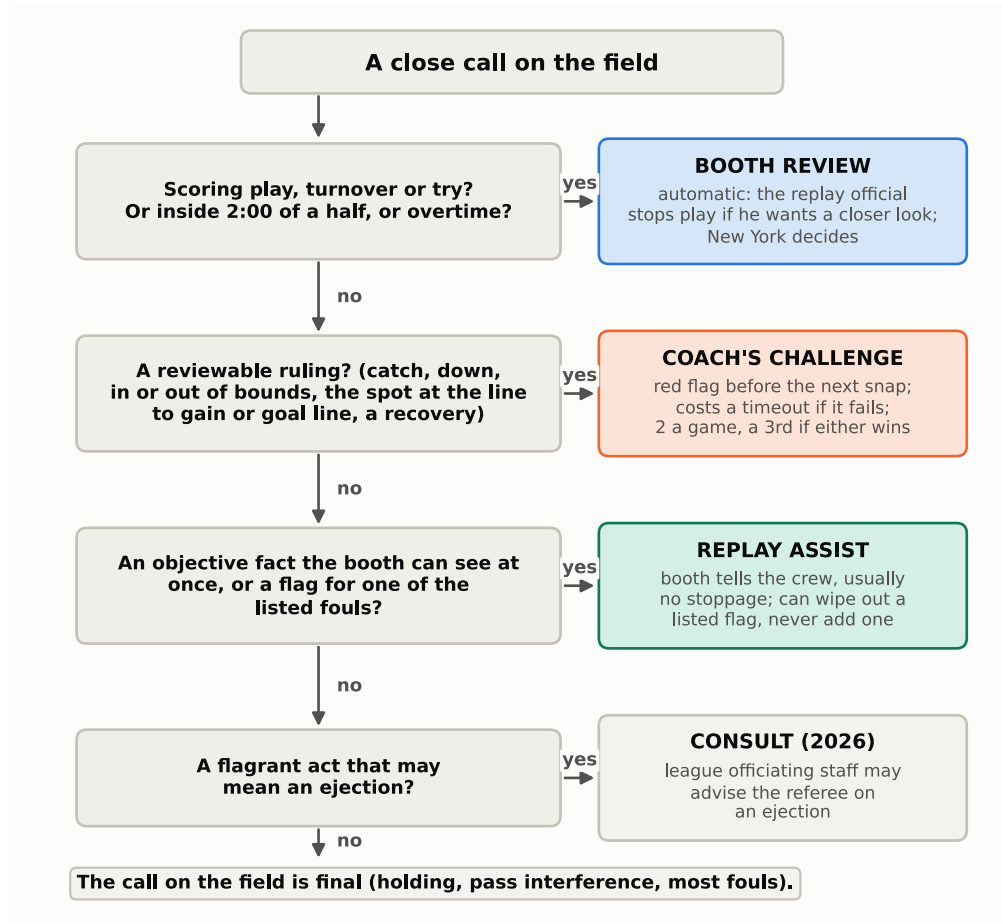


Figure 3: Who can review a close call, as of the 2026 season. Read from the top: plays the booth reviews automatically come first, and only plays that fall through to the second question can be challenged by a coach. Replay assist works alongside both, on clear objective facts and on flags for a short list of fouls, usually without stopping the game. Judgment fouls such as holding and pass interference can't be reviewed at all.

What actually gets reviewed

Play-by-play data records a review whenever one changes or stops the game, with the ruling that was reviewed and the result, so you can see where replay earns its keep.

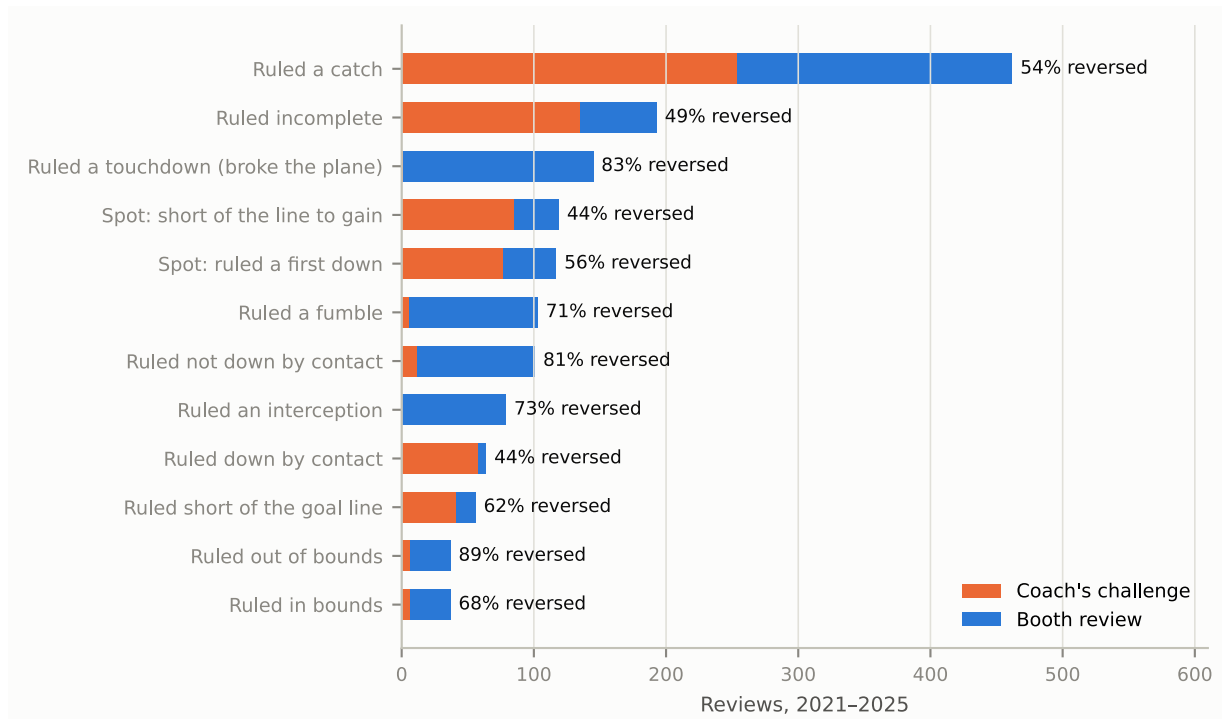


Figure 4: The twelve rulings reviewed most often, 2021–2025 (regular season and playoffs), split by who asked: a coach’s challenge (orange) or the booth (blue). The label gives the share of those reviews that reversed the call. Catch or no catch is by far the busiest question. Reversal rates look high for booth reviews such as ‘broke the plane’ because the play-by-play generally records a booth review only when it stopped the game, not the routine quick confirmations of every touchdown. Source: nflverse play-by-play (the review text in each play’s description).

Two patterns are worth carrying into a game. First, catch-or-no-catch rulings make up about 40% of all the recorded reviews, which is why the next section spends so long on the catch. Second, rulings of a *catch* are reviewed more than twice as often as rulings of *incomplete* (and overturned at a similar rate: 54% against 49%). Part of that is the booth looking at every touchdown catch. Part of it is by design: officials are taught that when in doubt they should let the play run and rule a catch or a fumble, because replay can take a catch away cleanly but can’t invent what would have happened after a whistle (more on that in a moment).

The catch

Nothing in the rulebook has caused more arguments than the definition of a catch, and the reason is the stakes. Whether a pass was caught decides more than one play. If it was a catch and the ball then comes loose, it’s a **fumble**: a live ball anyone can recover. If it wasn’t, it’s an **incomplete pass**: the ball is dead and the offense keeps it. A single definition separates “the offense keeps the ball and loses a down” from “the defense may have it”. So the rule has to draw a line between *possessing* a ball and merely *touching* it, at full speed, in the air, often while someone is hitting the receiver.

Today’s rule, rewritten in 2018, says a player has made a **catch** when, in bounds, he does three things in order:⁹

1. **Control.** He secures control of the ball in his hands or arms before it touches the ground.
2. **Feet (or a body part) down.** He touches the ground in bounds with both feet, or with any part of his body other than his hands (a knee, a hip, an elbow).
3. **Time.** After the first two, he makes an *act common to the game*: tucks the ball away, turns upfield, takes an extra step, reaches the ball toward the line, wards off a defender. Or he keeps control long enough that he *could* have.

Two clarifications do most of the work on replays. A ball that moves slightly in his hands while he keeps control is still controlled. And the ground can’t help him: if he goes to the ground before he has finished step 3, he must keep control when he lands. If the ball pops loose and touches the ground before step 3, the pass is incomplete; if it pops loose after step 3, it was a catch and then a fumble. An interception is judged by exactly the same three steps.

⁹NFL Rulebook (2026), Rule 8, Section 1, Article 3 (completed or intercepted pass): control in the hands or arms before the ball touches the ground; both feet or another body part other than the hands in bounds; then an act common to the game, or control long enough to perform one; Note 1, movement of the ball does not automatically mean loss of control; Note 2, a player who has not yet made the act and loses control when he hits the ground, in bounds or out, has not caught it: <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>.

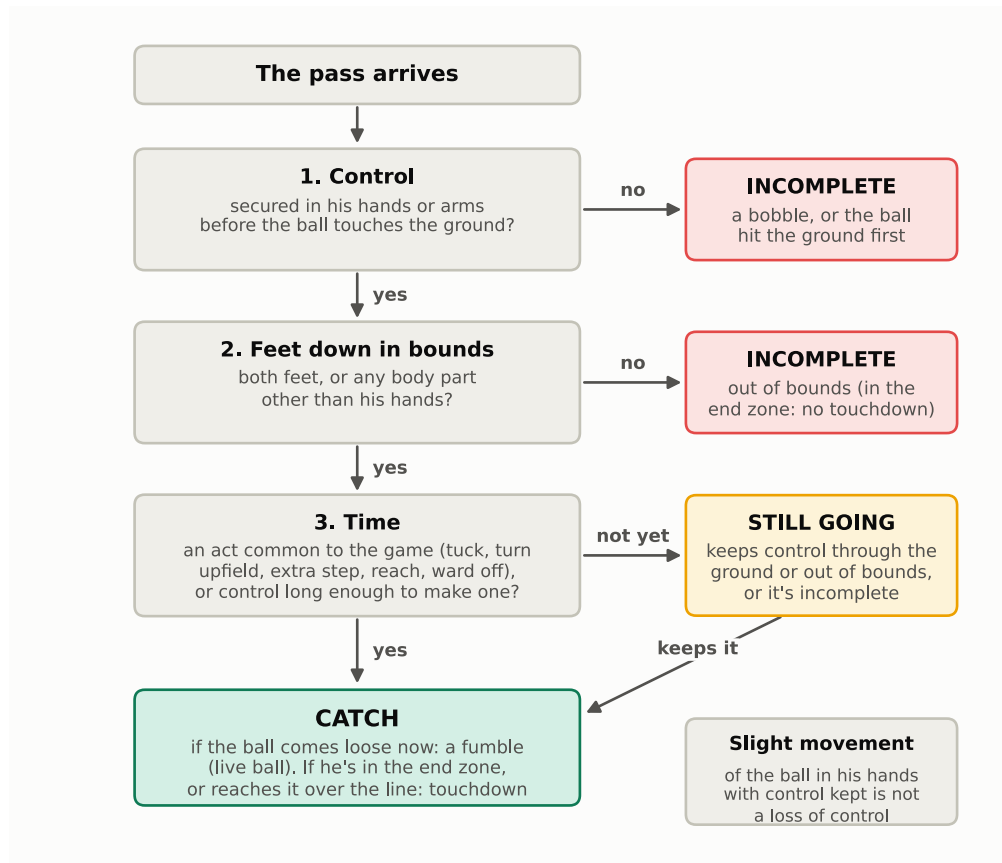


Figure 5: The catch rule as a replay official applies it (NFL rule since 2018). A pass must pass all three gates in order. Failing gate 1 or 2 means incomplete; the ball coming loose before gate 3 means incomplete; losing it after gate 3 means a catch and then a fumble. The same test decides an interception.

Why the rule was rewritten

For most of the 2010s the rule had an extra clause that produced some of the most hated rulings in league history. A receiver who went to the ground “in the process of the catch” had to keep control of the ball *after* hitting the ground, the so-called *survive the ground* requirement, no matter how many steps he had taken or how obviously he had possessed it. The clause was meant to make the rule objective: instead of asking officials to judge whether a player had possession long enough, it gave them a test they could see on video. The trouble was that the objective test kept ruling against what everyone watching had seen. Three plays made it untenable:¹⁰

- **Calvin Johnson, Lions at Bears, September 12, 2010.** Trailing 19–14 with 31 seconds left, Johnson caught a pass in the end zone, landed with both feet and his backside, and put the ball down on the turf as he rolled to get up. Ruled incomplete; the booth upheld it. The Lions lost by five, and “the Calvin Johnson rule” entered the vocabulary.
- **Dez Bryant, Cowboys at Packers, divisional playoffs, January 11, 2015.** Fourth-and-2 at the Green Bay 32, 4:42 left, Dallas trailing 26–21. Tony Romo threw deep down the left sideline; Bryant caught it, took three steps as he fell and reached the ball toward the goal line, and it touched the ground and moved as he landed at the 1. Ruled a catch on the field, reversed on Green Bay’s challenge. The Packers ran out the clock and won 26–21.
- **Jesse James, Patriots at Steelers, December 17, 2017.** With 34 seconds left and Pittsburgh down 27–24, James caught a pass at the 10, went to a knee, and lunged across the goal line; the ball moved as it hit the turf. Ruled a touchdown, reversed by the booth. The Steelers lost the game and, as it turned out, home-field advantage in the playoffs.

The league’s competition committee rewrote the rule that offseason, and the owners approved it 32–0 in March 2018. The new version deleted the going-to-the-ground clause: once a receiver has control and his feet down, a “football move” (a third step,

¹⁰All three plays appear in nflverse play-by-play: 2010 Week 1, Detroit at Chicago, 0:31 left in the fourth quarter (“pass incomplete deep right to 81-C.Johnson. The Replay Assistant challenged the incomplete pass ruling, and the play was Upheld”); the 2014 divisional round, Dallas at Green Bay, 4:42 left (“pass deep left to 88-D.Bryant to GB 1 for 31 yards ... Green Bay challenged the pass completion ruling, and the play was REVERSED”); 2017 Week 15, New England at Pittsburgh, 0:34 left (“pass short middle to 81-J.James for 10 yards, TOUCHDOWN. The Replay Official reviewed the pass completion ruling, and the play was REVERSED”). Final scores: Bears 19–14, Packers 26–21, Patriots 27–24.

reaching the ball toward the line, or simply time enough to make one) finishes the catch, and the rulebook spells it out as an “act common to the game”. The league’s football chiefs said publicly that under it, Bryant’s and James’s catches would have counted: reaching the ball toward the goal line is a football act.¹¹

The trade-off is worth naming, because it’s the same trade the whole replay system keeps making. The old rule was objective and produced absurd results; the new rule matches what people see, at the price of a judgment call (“did he have time?”) at the center of it. Disputes didn’t disappear; they moved from “did the ball move when he hit the ground?” to “was that lunge a football act or part of the fall?”

Common misconception

“The ball touched the ground, so it’s incomplete.” The ball is allowed to touch the ground during a catch, as long as the receiver keeps control of it, with his hands under it or the ball pinned to his body. What it can’t do is help him: a ball trapped against the turf with no hand under it, or one that pops loose when he lands before he has finished the catch, is incomplete.

Sidelines and end zones

Two places put extra weight on gate 2. At the **sideline**, a receiver needs both feet (or another body part) down in bounds with control; one foot, as in college football, isn’t enough. Since 2008 there has been no “force-out” ruling either: a receiver pushed out of bounds in the air before getting both feet down has not caught the ball, however sure the official is that he would have landed in bounds. (The one exception is a receiver who has control and is held up and *carried* out by a defender: that is still a catch.)¹² And because a toe-tap catch leaves no time for a football act before he goes out of bounds, he must keep control as he falls out; a ball jarred loose when he hits the ground out of bounds is incomplete.

In the **end zone**, the catch is the touchdown, so the receiver must complete all three steps there. That is different from a runner who already has the ball: for him, the ball only has to touch the plane of the goal line (the end zone is generous that way). Under the 2018 rule, a receiver who completes the catch short of the goal line and then reaches the ball across it has scored, which is exactly what Jesse James did.

Down, progress and the whistle

A play ends when the ball becomes dead, and most of the time that means a ball carrier going down. The NFL’s definition is different from the one you may know from college. A runner is **down by contact** when any part of his body other than his hands or feet touches the ground *after he has been touched by an opponent*. If nobody touched him, he isn’t down: a running back who slips on wet turf, or a returner who kneels to field a bouncing kick, may get up and keep running. In college football the runner is down as soon as his knee touches, contact or not.¹³ (A runner who slides feet-first or simply gives himself up is down where he does it; that’s how quarterbacks protect themselves.)

The order of events matters more than anything else on these plays. If the ball comes out *before* the knee (or elbow, or hip) touches after contact, it’s a fumble: live, and anyone’s. If the knee touches first, the runner was down and the ball coming out afterward doesn’t matter. On television these are separated by a few frames of video, which is why they’re among the most-reviewed rulings in the data above.

¹¹NFL.com, “NFL teams unanimously approve simplified catch rule” (Mar. 27, 2018; control, two feet or another body part down, and a football move such as a third step or reaching for the line, or the ability to perform one; the going-to-the-ground element removed): <https://www.nfl.com/news/story/0ap3000000923342/article/nfl-teams-unanimously-approve-simplified-catch-rule> ; CBS Sports, “NFL catch rule: Dez Bryant, Jesse James non-catches would now count” (Troy Vincent and Al Riveron on the two plays): <https://www.cbssports.com/nfl/news/nfl-catch-rule-dez-bryant-jesse-james-non-catches-would-now-count>.

¹²The “force-out” ruling, under which an official could rule a catch when a receiver was pushed out of bounds before landing, was eliminated for the 2008 season: NFL.com, “New NFL rules for 2008”: <https://www.nfl.com/news/new-nfl-rules-for-2008-09000d5d809f6279> . The carried-out exception: NFL Rulebook (2026), Rule 8, Section 1, Article 3, Note 5: <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>.

¹³NFL Rulebook (2026), Rule 7, Section 2, Article 1(a) (the ball is dead when a runner is contacted by an opponent and touches the ground with any part of his body other than his hands or feet) and 1(d) (a runner who slides or gives himself up): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook> ; NCAA Football Rules, Rule 4-1-3 (a ball carrier is down when any part of his body other than a hand or foot touches the ground, contact or not).

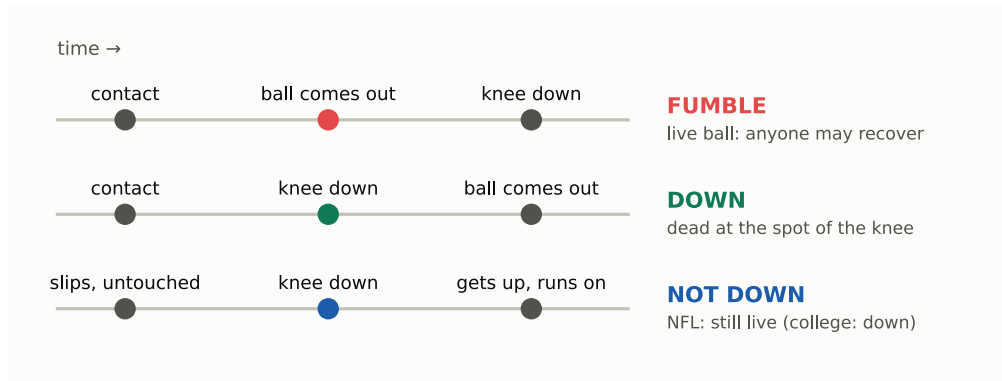


Figure 6: Down or fumble? It depends on the order of three events: the defender’s contact, the ball coming loose, and the runner’s knee (or any body part other than a hand or foot) touching the ground. Top: the ball comes out first, so it’s a fumble and the ball is live. Middle: the knee touches first, after contact, so he’s down and the play is dead before the ball comes loose. Bottom: nobody touched him, so in the NFL he isn’t down; he can get up and keep running (in college he’d be down).

Forward progress decides *where* the play ended. You met it in Downs, Distance, and Field Position: the ball is spotted at the farthest point it reached toward the goal line before the runner was stopped, not where he ended up after a pile drove him back. The same idea tells officials when to blow the whistle. A runner wrapped up and held upright, his legs still churning, has had his forward progress stopped even though he hasn’t gone down, and the officials will whistle the play dead rather than let a pile of defenders keep shoving him backward (or strip the ball from a man who has stopped moving forward). It also protects runners at the goal line: a back who gets the ball across the plane and is then knocked back into the field has scored. The protection covers only being *driven* back, though: a quarterback who retreats on his own and is sacked is spotted where he goes down, which is why a sack costs yards.

The whistle, and the 2008 lesson

The whistle is the one tool officials can’t take back. Once it blows, the play is dead, whatever happens next. That used to put replay in an impossible position: if an official ruled a play over (an incomplete pass, a runner down) and the video showed a fumble, replay could fix the ruling but could not know who would have recovered the ball if everyone had kept playing.

The play that made everyone face this came on September 14, 2008. Denver trailed San Diego 38–31 with 1:17 left, second-and-goal at the 1. Jay Cutler dropped back, and the ball slipped out of his hand as he started to throw; a Charger fell on it. Referee Ed Hochuli ruled it an incomplete pass and blew his whistle. Replay showed a fumble, but because the whistle had already killed the play, the Chargers couldn’t be given the ball. Denver kept it at the 10, where Cutler had lost it, scored on fourth-and-goal with 29 seconds left, went for two, and won 39–38.¹⁴ Hochuli, one of the most respected referees of his era, publicly apologized.

Two things changed. In 2009 the owners passed what was quickly nicknamed the “Hochuli rule”: replay may now award the ball to the team that recovered it when the recovery is clear and in the *immediate continuing action*, even after a whistle.¹⁵ And the officials’ habit changed: on a bang-bang fumble, they are taught to hold the whistle, let the players finish the scramble for the ball, and rule a fumble when in doubt, because replay can turn a fumble back into a dead ball cleanly but can’t recreate a recovery that never happened. That is why you now see plays continue for a few seconds after a runner seems to be down, and a defender scooping up a loose ball and running the other way while everyone else has stopped.

Pass or fumble: the quarterback’s arm

A close cousin of down-or-fumble is pass-or-fumble, and it comes up every time a pass rusher hits a quarterback as he throws. The rule is about the arm. Once the quarterback, holding the ball, starts his arm moving forward to throw, anything that comes out is a forward pass, and if it hits the ground it’s incomplete. If he loses the ball *before* his arm starts forward, or while he is pulling it back down to tuck it after deciding not to throw, it’s a fumble.

The second half of that sentence was not always true. Until 2013 the rulebook had the **tuck rule**: if the quarterback started forward and then brought the ball back toward his body, it was still a pass until he had tucked it all the way in. It is remembered for

¹⁴nflverse play-by-play, 2008 Week 2, San Diego at Denver, 1:17 left in the fourth quarter, second-and-goal at the 1: “6-J.Cutler pass incomplete short right to 28-M.Pittman. The Replay Assistant challenged the fumble ruling, and the play was REVERSED. ... FUMBLES (Team), declared dead at LAC 10.” Final score Broncos 39, Chargers 38. On Hochuli’s apology: “Ed Hochuli,” Wikipedia: https://en.wikipedia.org/wiki/Ed_Hochuli.

¹⁵ProFootballTalk, “Hochuli rule approved” (March 2009): <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/hochuli-rule-approved>. The current wording (“possession may be awarded to a player who clearly recovers a loose ball in the immediate continuing action”): NFL Rulebook (2026), Rule 15, Section 2, Article 3: <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>.

one snowy night in Foxborough: the divisional playoff in January 2002, when Tom Brady lost the ball under a hit from Charles Woodson late in the fourth quarter, the Raiders recovered, and replay applied the tuck rule to call it an incomplete pass. New England kept the ball, tied the game with a field goal and won in overtime, the first playoff run of a dynasty. The league abolished the rule in 2013.¹⁶ Today the same play is a fumble.

Because these are turnovers when ruled fumbles, the booth reviews every one of them. When you see a strip-sack ruled on the field, watch the arm, not the ball: if the arm was coming forward with the ball in the hand, expect “reversed, incomplete pass”.

A fumble through the end zone

Now the harshest rule in the book. When a fumble goes out of bounds, the general rule is generous to the team that fumbled: it keeps the ball. If the ball went out going backward, it’s spotted where it went out; if it went out going *forward*, it comes back to the spot of the fumble, so that nobody gains yards by fumbling forward on purpose.

The exception is the opponent’s end zone. A **fumble through the end zone**, a ball fumbled forward into the end zone that goes out of bounds there, out the side or out the back, is a **touchback**: the defense gets the ball at its own 20, even though no defender ever touched it.¹⁷

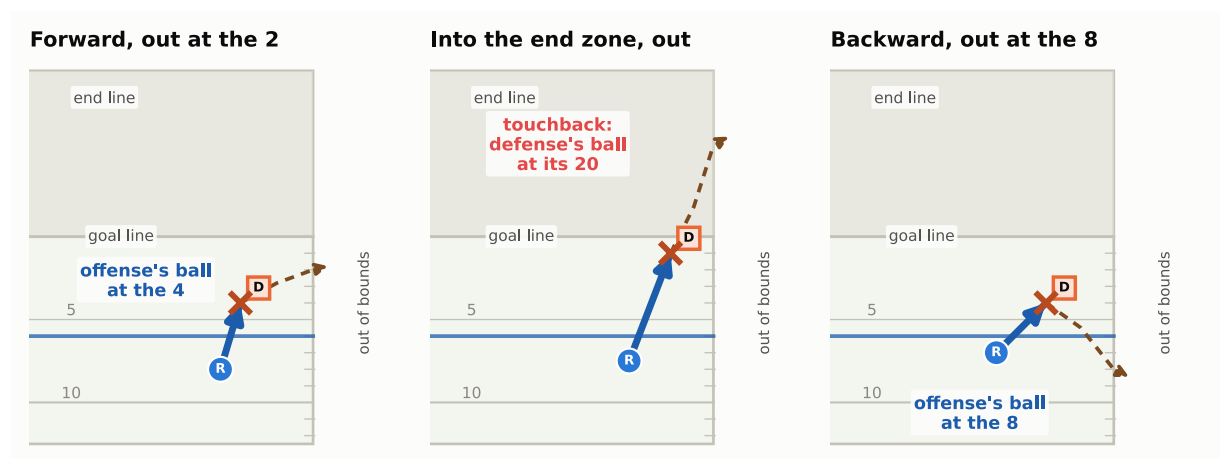


Figure 7: Three fumbles out of bounds near the goal line, offense moving up the page. Left: the ball squirts forward and out at the 2: the offense keeps it, but back at the 4, the spot of the fumble, because a forward fumble can’t gain ground. Middle: knocked loose at the 1, it bounces into the end zone and out the side: a touchback, and the defense takes over at its own 20. Right: the ball goes backward and out at the 8: the offense keeps it where it went out. The X marks the fumble, the dashed brown line the ball’s path.

Why would the rules punish the offense so hard? The logic is about territory, not possession. The end zone belongs to the defending team. When a ball goes dead in or behind your end zone because of something the other team did (a punt rolls through, a kickoff sails out the back), the result is a touchback, your ball at your 20, like the touchbacks on punts. A fumble into your end zone and out is treated the same way: the offense put the ball there, it died there, and it’s yours. The rule is consistent; it just doesn’t feel fair, because the offense had the ball a moment earlier and the defense did nothing to earn it. It is one of the most criticized rules in the book, and ideas to soften it (giving the ball back to the offense at the spot of the fumble, say) come up every few years. The league’s competition committee looked at it and kept it in 2017 and again in February 2024, after a playoff fumble through the end zone, arguing that the ball carrier is responsible for the ball and noting that nobody had come up with a workable alternative. It is unchanged in the 2026 rulebook.¹⁸

The rule shapes technique. It’s why coaches drill runners and receivers to carry the ball in the arm away from the defenders near the goal line, why a ball carrier stretching for the pylon is taking a real gamble, and why defenders near the goal line punch at the

¹⁶“Tuck Rule Game,” Wikipedia (the January 19, 2002, divisional playoff between the Raiders and Patriots, and the rule’s abolition in 2013): https://en.wikipedia.org/wiki/Tuck_Rule_Game.

¹⁷NFL Rulebook (2026), Rule 8, Section 7, Article 3, Items 3 and 4 (a fumble out of bounds forward is returned to the spot of the fumble; one that goes forward into the opponent’s end zone and over the end line or sideline is a touchback for the defense) and Rule 11, Section 6 (touchbacks): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>. The Seferian-Jenkins play is in nflverse play-by-play, 2017 Week 6, New England at the Jets, 8:31 left in the fourth quarter (“The Replay Official reviewed the runner broke the plane ruling, and the play was REVERSED ... FUMBLES (21-M.Butler), ball out of bounds in End Zone, Touchback”); final score Patriots 24, Jets 17.

¹⁸CBS Sports, “NFL competition committee sees no need to change rule on fumbling through end zone, per report” (Feb. 26, 2024; the second time since 2017 the committee declined to change it): <https://www.cbssports.com/nfl/news/nfl-competition-committee-sees-no-need-to-change-rule-on-fumbling-through-end-zone-per-report/>. Rule 8, Section 7, Article 3, Item 4(a) of the 2026 rulebook still makes it a touchback.

ball instead of wrapping up. A clean tackle at the 1 gives the offense first-and-goal; a punched ball through the end zone gives the defense the ball at the 20.

! Why it exists: the fourth-down fumble rule

One more fumble rule has a famous origin. On September 10, 1978, with the Raiders trailing the Chargers in the final seconds, Ken Stabler, about to be sacked, fumbled the ball forward; teammates batted and kicked it toward the goal line, and Dave Casper fell on it in the end zone for the winning touchdown. The “Holy Roller” was legal then. The next season the league made it impossible: on fourth down, and on any down after the two-minute warning, only the player who fumbled may recover and advance the ball. If a teammate recovers it, the ball comes back to the spot of the fumble (or the spot of recovery, if that is behind it).¹⁹ The rule exists to stop an offense from fumbling forward on purpose when it has nothing left to lose.

Backward passes and forward passes

Football allows exactly **one forward pass per play**, and it must be thrown from behind the line of scrimmage. It allows any number of **backward passes** (also called **laterals**), from anywhere, by anyone, at any time. The two are treated very differently when they hit the ground:

- A **forward pass** that hits the ground is **incomplete**. The ball is dead, and the offense keeps it.
- A **backward pass** that hits the ground is **still live**, exactly like a fumble. Either team may recover it, and either team may advance it.

Which is which? The test is the ball’s direction: compare the yard line where it left the passer’s hand with the yard line where it is first touched (or first hits the ground). If the second is closer to the opponent’s goal, the pass was forward. If it’s the same yard line or behind, the pass was backward, so a pass thrown perfectly parallel to the line counts as **backward**.²⁰ Notice that it’s the ball’s path that counts, not the players’ positions on the field.

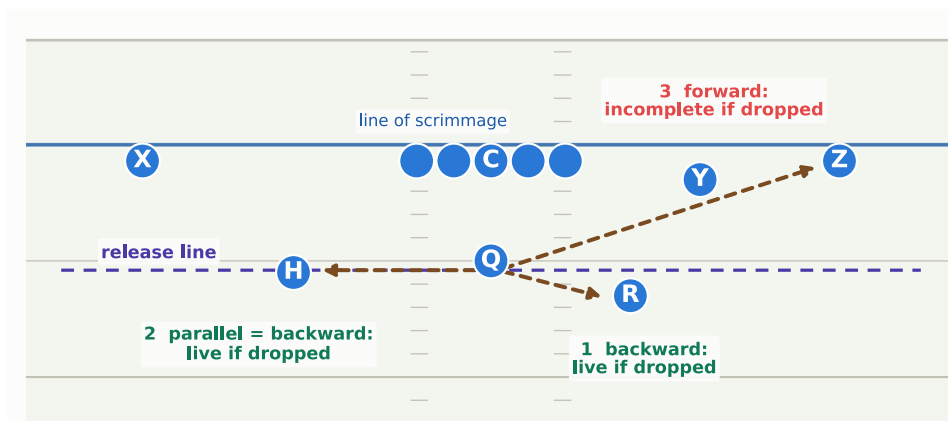


Figure 8: Forward or backward is decided by the yard line where the ball leaves the passer’s hand (dashed ‘release line’) and the yard line where it’s first touched. From the shotgun, the quarterback catches the snap five yards deep. A pitch to the running back (1) goes behind the release line: a backward pass, live if it hits the ground. A bubble screen (2) caught after the slot receiver has bowed back to the release line is parallel, which counts as backward. A quick screen to the outside receiver (3) is caught four yards ahead of the release line: a forward pass, incomplete (and dead) if it’s dropped.

Why does a dropped backward pass stay live? Because otherwise the backward pass would be a free way to move the ball sideways with no risk, and the rules want risk to come with that freedom. The result shows up in three places on a broadcast:

¹⁹“Holy Roller (American football),” Wikipedia (Raiders at Chargers, September 10, 1978, and the 1979 rule limiting fourth-down and late-game fumble advances to the fumbling player): [https://en.wikipedia.org/wiki/Holy_Roller_\(American_football\)](https://en.wikipedia.org/wiki/Holy_Roller_(American_football)); today’s wording: NFL Rulebook (2026), Rule 8, Section 7, Articles 5 and 6.

²⁰NFL Rulebook (2026), Rule 8, Section 1, Articles 1 and 2 (a forward pass and the one legal forward pass per down, thrown from behind the line); Rule 3, Section 12, Article 2 (a pass parallel to a yard line is backward); Rule 8, Section 7, Article 1 (a runner may throw a backward pass at any time, and players of either team may advance after catching it or recovering it after it touches the ground): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>.

- **Pitches and option plays.** A pitch on the option is a backward pass. Drop it and you’ve fumbled, in effect, which is why option teams drill the pitch relentlessly.
- **Screens.** Most screens thrown from the shotgun are forward passes, because the quarterback releases the ball five or six yards behind the line and the receiver catches it closer to the line than that. But a bubble or swing screen to a player who has drifted back level with the quarterback may be backward, and a defender who sees it hit the ground should pick it up and run. Coaches teach defenders to scoop up every loose screen and let the officials sort it out. The rules on linemen downfield follow the same line: the illegal-man-downfield rule applies only when a forward pass is thrown beyond the line, so on a true backward pass linemen may already be blocking downfield.
- **Last-play desperation.** The multi-lateral kickoff return or final snap, the ball pitched backward again and again, is legal because every pitch is backward.

The most famous backward pass in NFL history came in a wild-card game on January 8, 2000. Buffalo had just kicked a go-ahead field goal with 16 seconds left. On the kickoff, Tennessee’s Lorenzo Neal handed the ball to tight end Frank Wycheck, who turned and threw it across the field to Kevin Dyson, who ran 75 yards untouched down the sideline for the winning touchdown. Replay looked at whether Wycheck’s throw had gone forward. It hadn’t been shown to: the ruling of a legal lateral stood, and the “Music City Miracle” became the standard illustration of the rule that what counts is the ball’s path from hand to hands.²¹

Intentional grounding and the tackle box

Quarterback Play introduced the throwaway and the rule that limits it. Here is the full rule, and the reasoning behind it.

Intentional grounding is a foul when a passer who is facing an imminent loss of yardage because of the rush throws a pass that isn’t in the direction and vicinity of an originally eligible receiver. Three conditions, all needed:²²

1. **Pressure.** He must be facing a likely sack. A quarterback under no pressure who throws the ball into the turf has wasted a down, but he hasn’t grounded it.
2. **No realistic target.** The pass isn’t thrown near an eligible receiver. “Near” is a judgment, and throwing the ball well over the head of a covered receiver in the right part of the field counts.
3. **Inside the tackle box.** The **tackle box**, which the rulebook calls the *pocket area*, is the region between the outside edges of where the two offensive tackles lined up, running from the line of scrimmage all the way back to the offense’s own end line.

The exception is what makes the box matter: **a passer who is or has been outside the tackle box may throw the ball away freely, as long as it lands at or beyond the line of scrimmage** (extended to the sidelines), including out of bounds past it. Since the 2024 rulebook he counts as outside the box if any part of his body, or the ball, is outside it. And “has been” means what it says: a quarterback who escapes the box and then drifts back inside it keeps the exception for the rest of the play.

The penalty is built to cost about what the sack would have: **loss of down and 10 yards** from the previous spot, or loss of down at the spot of the pass if that is farther back, and **a safety** if he throws it from his own end zone. The other exception you’ll see every week is the **spike**: a quarterback who throws the ball straight into the ground immediately after taking the snap, to stop the clock, isn’t penalized; it costs a down and nothing else. Spikes belong to the end-of-half clock game, which is in The Clock, the Two-Minute Drill, and Fourth Down.

²¹nflverse play-by-play, 1999 wild-card round, Buffalo at Tennessee, January 8, 2000, 0:16 left: “S.Christie kicks 46 yards from BUF 30 to TEN 24. L.Neal to TEN 25 for 1 yard. Lateral to F.Wycheck to TEN 25 for no gain. Lateral to K.Dyson for 75 yards, TOUCHDOWN.” “Music City Miracle,” Wikipedia (the replay review and the 22–16 final): https://en.wikipedia.org/wiki/Music_City_Miracle.

²²NFL Rulebook (2026), Rule 8, Section 2, Article 1 (intentional grounding: the pass must be in the direction and vicinity of an originally eligible receiver unless a passer who is or has been outside the pocket area throws it to land at or beyond the line of scrimmage extended; penalty loss of down and 10 yards or at the spot of the pass, a safety from the end zone; the spike exception) and Rule 3, Section 25 (the pocket area, between the outside edges of the normal tackle positions, back to the end line): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>. On the 2024 wording (any part of the body or the ball outside the pocket): Mike Florio, “New rule book subtly makes it easier for quarterbacks to throw ball away,” ProFootballTalk, July 31, 2024: <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/new-rule-book-subtly-makes-it-easier-for-quarterbacks-to-throw-ball-away>.

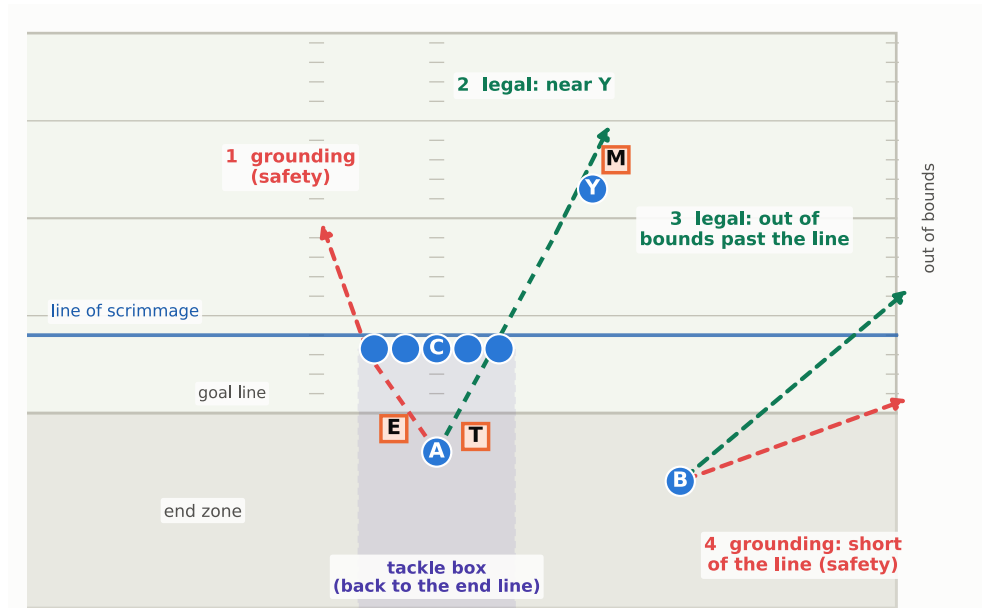


Figure 9: Intentional grounding backed up at the offense’s own 4, where the tackle box (shaded) runs back into the end zone. Quarterback A, inside the box and in his end zone, throws to an empty part of the field (1): grounding, and from the end zone that’s a safety. Throwing it over the covered tight end’s head instead (2) is a legal throwaway. Quarterback B has escaped the box: throwing it out of bounds past the line of scrimmage (3) is legal even though no one is near, but a throw that lands out of bounds short of the line (4) is grounding, also a safety from the end zone.

Why draw the line at the box? The rule is balancing two things. A sack is something the defense earned: it beat the blockers. If a quarterback could always throw the ball into the ground to avoid one, the pass rush would lose much of its value, so inside the pocket, where the defense has him trapped, he must at least throw toward someone. But a quarterback who has escaped the pocket has beaten the rush himself, and he’s usually near the sideline with defenders closing; letting him throw it away (provided it reaches the line, so he can’t just dump it at his feet) rewards the escape and spares him a hit. The 2024 wording, counting the ball or any body part outside the box, tilted that balance a little further toward the quarterback.

The best-known grounding call of recent decades shows the safety clause at work: on New England’s first offensive play of Super Bowl XLVI, February 5, 2012, Tom Brady, pressured in his own end zone, threw deep down the middle to nobody. Grounding from the end zone is a safety, and the Giants led 2–0 before the Patriots had gained a yard. They won by four.²³

The “has been outside” clause is the subtle part, and it’s easiest to see in motion.

²³nflverse play-by-play, Super Bowl XLVI (2011 season), New York Giants vs New England, February 5, 2012, first quarter, 9:00, first-and-10 at the New England 6: “12-T.Brady pass incomplete deep middle [91-J.Tuck]. PENALTY on NE-12-T.Brady, Intentional Grounding, 6 yards, enforced in End Zone, SAFETY.” The Giants won 21–17.

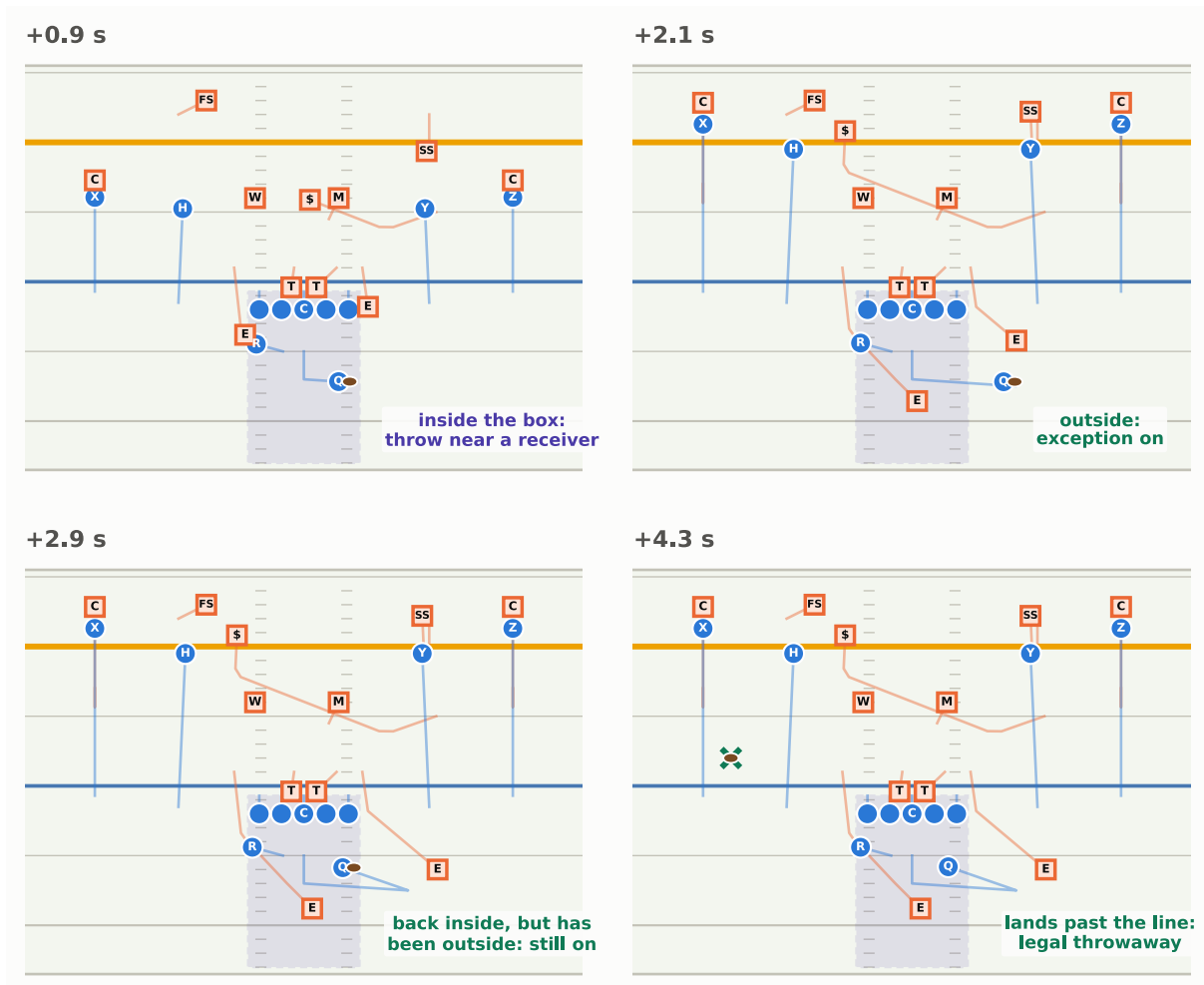


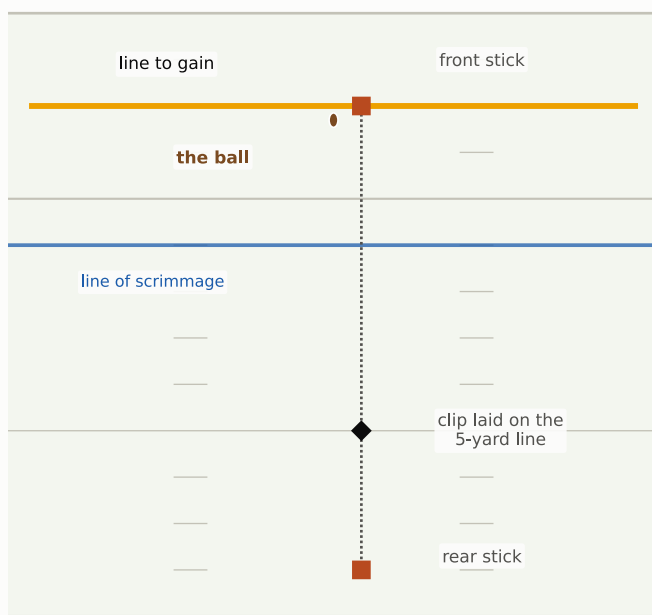
Figure 10: Once outside, always outside (simulated tracking in Big Data Bowl format; an illustrative play). 0.9 s: the quarterback is in the tackle box (shaded), so a throwaway must go near a receiver. 2.1 s: pressure flushes him to his right, out of the box; the grounding exception is now on. 2.9 s: he drifts back inside the box, but he has been outside, so the exception stays on for the rest of the play. 4.3 s: he throws it away into empty space on the left, and it lands past the line of scrimmage: legal, no flag. Had it landed short of the line, it would be grounding.

The measurement: chains, cameras, and the spot

Downs, Distance, and Field Position described the chain crew and its century-old ritual: two sticks joined by ten yards of chain, carried onto the field and stretched out to see whether the ball's nose had reached the front stick. Since 2025 that walk is mostly gone. A **virtual measurement**, often called a **Hawk-Eye measurement** after the Sony system that does it, uses six 8K cameras around the stadium to measure the distance from where the ball sits to the line to gain, and hands the answer to the officials in about 30 seconds, while a virtual graphic of the measurement plays on the broadcast and the stadium screens. The chain crew stays on the sideline as the visible marker for players and coaches, and as a backup if the cameras fail.²⁴

²⁴NPR, "Say goodbye to chain crews: The NFL will use camera technology to measure 1st downs" (Apr. 1, 2025): <https://www.npr.org/2025/04/01/nx-s1-5347990/nfl-first-downs-measurements-hawk-eye> ; Sony, "Sony's Hawk-Eye Innovations selected by the NFL to revolutionize line-to-gain measurements": <https://sony.mediaroom.com/2025-04-01-Sonys-Hawk-Eye-Innovations-Selected-by-the-NFL-to-Revolutionize-Line-to-Gain-Measurements> ; CBS Sports on the 2025 debut (six 8K cameras, about 30 seconds per measurement): <https://www.cbssports.com/nfl/news/no-chain-no-problem-nfl-debuts-new-virtual-measurement-system-during-hall-of-fame-game>.

Until 2024: the chains walk out



Since 2025: cameras measure the gap

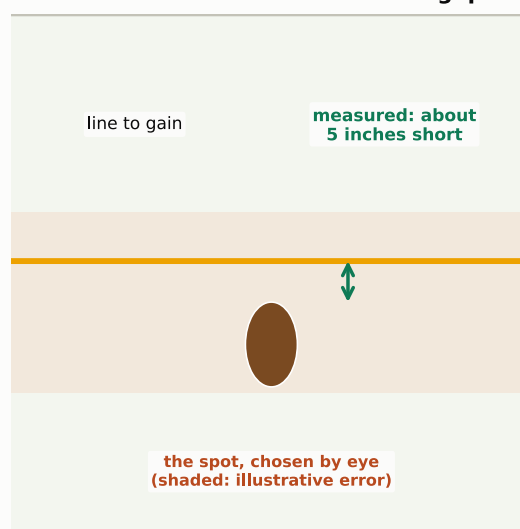


Figure 11: What changed in 2025, and what didn't. Left, the old way: the chain crew carries the sticks onto the field, an official lays the chain's clip on the 5-yard line where it was marked, so the rear stick goes back exactly where the series began, and the front stick is compared with the nose of the ball. Right, a close-up of the same spot, about two yards across: since 2025, six cameras around the stadium measure the gap between the ball's nose and the line to gain, to a fraction of an inch. But in both, the ball sits where an official spotted it by eye (the shaded band is an illustrative margin of error, not a measured one): the cameras measure from that spot; they don't choose it.

The change is a nice lesson in the difference between *precision* and *accuracy*. The chains were neither precise (a clip laid by hand, a chain pulled taut by eye) nor fast. The cameras are extremely precise. But they measure from the spot, and the spot is still a human judgment: an official running alongside the play, often looking through a pile of bodies, deciding where the ball's nose was when forward progress stopped. A measurement accurate to a fraction of an inch, taken from a spot that could be off by a foot, is only as good as the spot. That is why the spot itself can still be reviewed, but only in narrow cases: on a play that's challenged or booth-reviewed, or through replay assist, the booth can move the spot when the video clearly shows where the ball was relative to the line to gain or the goal line.

Common misconception

"The cameras decide first downs now." They decide the *measurement*. The officials still spot the ball by eye, and the yellow line on your screen is still the broadcaster's drawing, not an official mark. When a spot looks wrong, the cameras can't help; only the replay booth's look at the spot can.

Reading the referee's announcement

Every announcement follows the same grammar, and once you know it you can usually finish the referee's sentence for him. For a foul, the order is: **what** the foul was, **who** committed it (offense or defense, and the player's number), **how much** it costs and **from where**, and **what happens to the down**. For a review it's: the **outcome** (stands, confirmed, reversed), the **reason** in a sentence, and the **result** (the new down, distance and spot, and anything about the clock).

Table 2: Six announcements decoded. The words in quotation marks are the typical form; referees vary the wording a little.

What the referee says	What it means
“Holding, number 72, offense. Ten-yard penalty. Repeat second down.”	An accepted offensive holding foul: 10 yards from the previous spot, and the down is played again.
“Pass interference, defense, number 24. Spot foul. Automatic first down.”	Defensive pass interference: the ball goes to the spot of the foul with a new set of downs.
“Offsetting fouls. Repeat third down.”	A foul by each team; they cancel and the down is replayed.
“Illegal formation, offense. The penalty is declined. Fourth down.”	The defense chose the play’s result over the yards (a third-down incomplection, say).
“After review, the ruling on the field stands. Third down.”	The video wasn’t clear enough to overturn the call, whichever way it was.
“After review, the ruling on the field is reversed. The runner’s knee was down before the ball came out. There is no fumble. Second down.”	The booth turned a turnover back into a tackle: the down-by-contact order of events.

The enforcement words in those sentences, *spot foul*, *declined*, *automatic first down* and the rest, are taught in Contact Rules and Penalties, which also explains when a team should decline a penalty. Before the announcement, the referee’s arm signal tells you most of it, and a few signals cover most flags: a hand gripping the other wrist for holding; both hands on the hips for offside or encroachment; forearms rolling over each other in front of the chest for a false start; both open hands pushed forward from the shoulders for pass interference; one wrist chopping the other above the head for a personal foul; and both arms, held parallel, waved diagonally across the body for intentional grounding.²⁵

Do crews call games differently?

Each referee leads the same crew for a season, and fans and bettors love the idea that some crews are “flag-happy”. The data gives a measured answer.

²⁵NFL Rulebook (2026), “Official Signals” (holding: grasping one wrist, the fist clenched, in front of the chest; offside or encroachment: hands on hips; false start: forearms rotated over and over in front of the body; pass interference: hands open and extended forward from the shoulders; personal foul: one wrist striking the other above the head; intentional grounding: parallel arms waved in a diagonal plane across the body): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>.

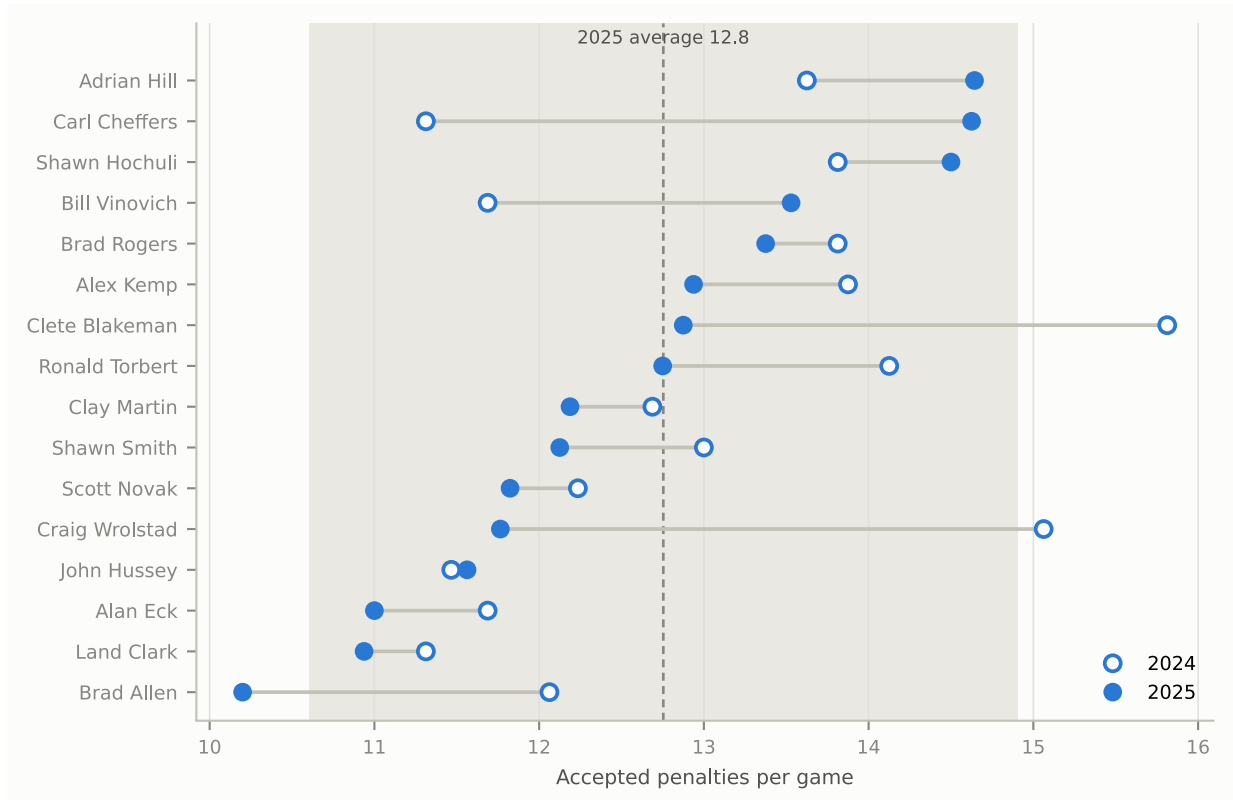


Figure 12: Accepted penalties per game (both teams combined, every play) in games led by each referee’s crew, 2024 (hollow) and 2025 (filled), for the referees who led a crew in both seasons, sorted by 2025. The grey band is the range around the 2025 league average that a crew would land in from game-to-game randomness alone, about two standard errors for a 16-game sample. Most crews sit inside it, and the order reshuffles from one season to the next. Source: nflverse play-by-play and officials data (regular seasons).

In 2025 an average game had 12.8 accepted penalties, and the crews’ averages ran from 10.2 (Brad Allen’s crew) to 14.6 (Adrian Hill’s).²⁶ That spread sounds large until you set it against the noise. A single game’s total varies a lot (the standard deviation was about 4.3 penalties), so a crew’s 16- or 17-game average would wander about ± 2.2 around the league average by luck alone. Most crews sit inside that band. And the ranking doesn’t hold: the correlation between a crew’s 2024 and 2025 averages was 0.28, and across the ten seasons from 2016 to 2025 the year-to-year correlation ranged from -0.35 to 0.76 , with a median of 0.28 . Only 12 of the 16 crews landed in the same half of the table both years.

So crews differ a little, and inconsistently. Most of what makes a game flag-heavy is the game: the teams (some offenses false-start more, some defenses hold more), the noise, the weather, the number of plays. And some of the variation is the league’s doing, not the crews’: the league-wide average swings with points of emphasis, from 11.2 to 13.5 per game across 2016–2025, which is why a preseason memo telling officials to watch for a particular foul can move the numbers for every crew at once.

²⁶Course calculation, 2024 and 2025 regular seasons: nflverse `load_officials()` (the referee of each game) joined to play-by-play; accepted fouls counted from the play text, where nflverse writes an accepted foul as “PENALTY on” and a declined or offsetting one as “Penalty on”. Both teams’ fouls on every play, including kicks and pre-snap fouls, are counted. The band uses the 2025 game-level standard deviation and the average number of games per crew. Year-to-year correlations use every referee who led a crew in consecutive seasons, 2016–2025.

Film room

i Film room: Cowboys at Packers, 2014 divisional playoffs, the catch that wasn't

Find the play: January 11, 2015, fourth quarter, 4:42 left, fourth-and-2 at the Green Bay 32, Dallas down 26–21. Watch Dez Bryant from the moment the ball arrives, and count the yards. Control: yes, he has it in his hands above the cornerback. Feet: yes, he lands and takes steps. Time: here is the argument. He takes three steps while falling and stretches the ball toward the goal line, and the ball hits the turf and moves as he lands. Under the 2014 rule, he was going to the ground in the process of the catch, so he had to keep control through the ground, and replay ruled he hadn't. Under the 2018 rule, the reach is an act common to the game, which finishes the catch before the ball touches down. Watch the replay twice, once with each rule in your head, and you'll understand why the league changed it.

i Film room: Patriots at Steelers, 2017, survive the ground

December 17, 2017, 34 seconds left, Steelers down 27–24, first-and-10 at the New England 10. Jesse James catches a short pass, goes to a knee at about the 1, and lunges across the goal line; the ball moves as it hits the turf. Notice first who reviewed it: under two minutes, and a scoring play, so it was the booth, not a challenge. Then notice the clause that decided it, the same going-to-the-ground language as the Bryant play. Two plays later Ben Roethlisberger faked a spike and threw an interception in the end zone. Three months later the rule was gone.

i Film room: Patriots at Jets, 2017, the touchback nobody recovered

October 15, 2017, fourth quarter, 8:31 left, the Jets down 24–14 at the New England 4. Austin Seferian-Jenkins catches a short pass and turns toward the goal line; as he dives for it, with a defender on him, the ball comes loose. Ruled a touchdown, it went to the booth as a scoring play. Replay ruled he lost the ball before crossing the line, and that it then went out of bounds in the end zone. Watch the referee's announcement: no defender recovered anything, and the Patriots got the ball at their own 20. New England won by seven. Then watch the ball as he dives: anything held away from the body near the goal line is exposed to exactly this outcome.

i Film room: Chargers at Broncos, 2008, the whistle

September 14, 2008, 1:17 left, Denver down 38–31, second-and-goal at the San Diego 1. Watch Jay Cutler's arm. It never comes forward with the ball; the ball slips out as he winds up. Then listen for the whistle, and watch the Chargers stop and start, unsure whether the play is alive. The rules of the time gave replay no way to award the recovery after the whistle. Today's officials would let the scramble finish, rule a fumble, and let replay confirm it.

i Film room: the 2025 season, the cameras arrive

Pick any 2025 or 2026 game and wait for a close spot on third or fourth down. Watch the order of events: the official spots the ball by eye, the referee signals for a measurement, and within about half a minute the stadium screen shows the virtual line against the ball's nose. The chain crew never leaves the sideline. Then watch the reactions: players and coaches now argue about the spot, the only human part left, and almost never about the measurement.

Watch for it on Sunday

💡 Watch for it: the ref's mic

Every time a flag flies, or a ruling is close, make two calls before the referee turns on his microphone:

1. **Name the official.** Who threw it? A flag from the line of scrimmage at a sideline is the down or line judge (pre-snap foul, likely false start, offside or illegal formation). A flag from deep downfield is a side, field or back judge (likely pass interference or illegal contact). A flag from the backfield, near the quarterback, is the referee (roughing the passer, grounding) or the umpire (holding in the middle).
2. **Name the foul or the ruling,** and its result: how many yards, from where, and the down that follows.
3. **On every review, predict the outcome:** *reversed*, *confirmed* or *stands*. First ask who started it (a coach's flag, or the booth: under two minutes, a score, a turnover). Then ask which angle would settle it. If the decisive moment is hidden (a pile, a body in the way), say *stands*.
4. **On every close catch, count the gates out loud:** control, feet, time.

Keep a tally for one game: officials named correctly, fouls named correctly, review outcomes called correctly. You'll miss plenty at first; by the end of a season the official-then-foul habit will have you announcing it before the referee does. Add your review predictions to your prediction log.

Predict the play

💡 Predict the play 1: the sideline catch

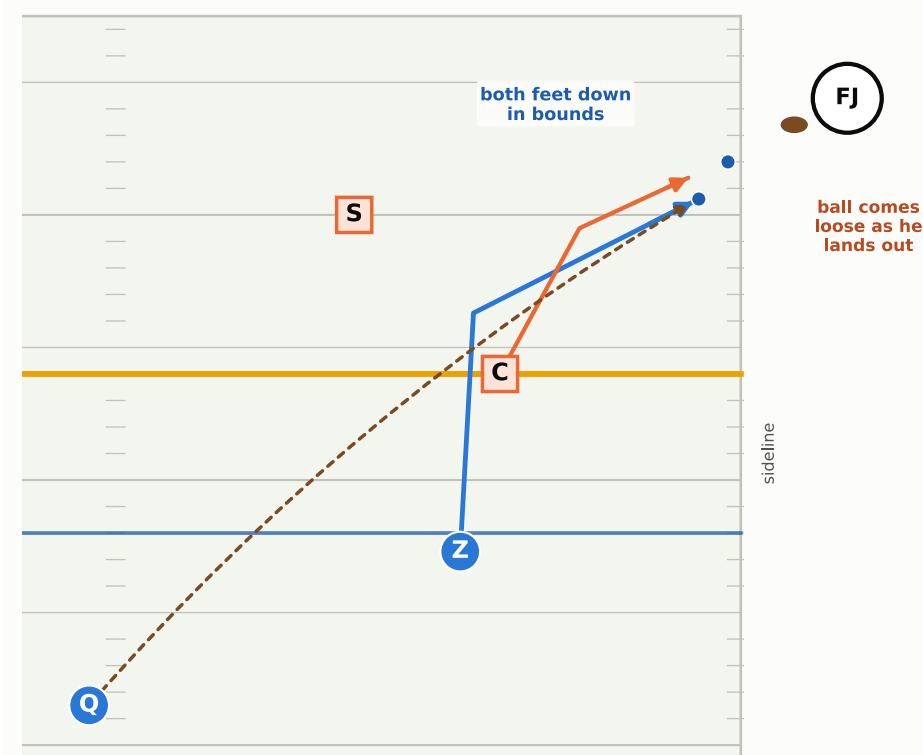


Figure 13: Second quarter, 1:52 left. The receiver (Z) catches a pass running toward the right sideline at the opponent's 38, gets his right foot down in bounds, then his left, and is knocked out of bounds by the cornerback; the ball comes loose as he hits the ground beyond the sideline. The field judge (FJ) rules it incomplete. The offense's head coach reaches for his red flag.

Can the coach challenge? Who decides, and what will the review look for?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: reaching for the pylon

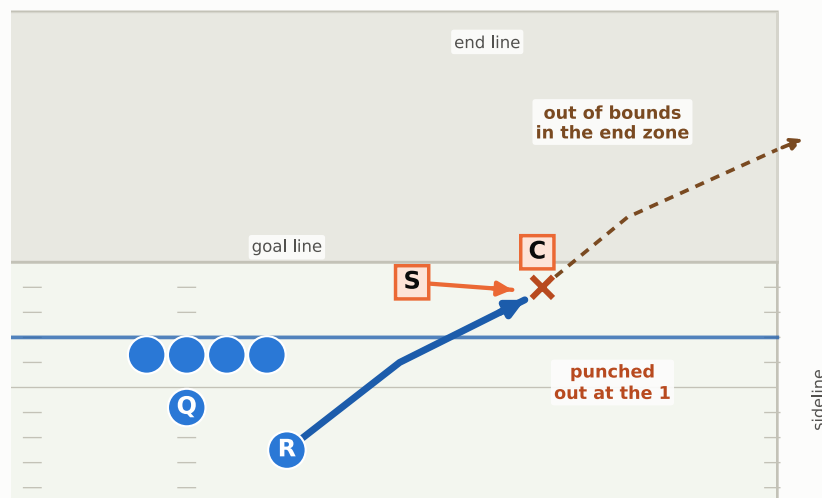


Figure 14: Third-and-goal from the 3, third quarter. The running back (R) bounces outside and dives for the pylon. At the 1, a safety's punch knocks the ball out of his outside arm; it bounces into the end zone and rolls out of bounds through the side of the end zone. Nobody touches it after the punch.

What's the ruling, and who gets the ball where? What if the ball had gone out of bounds at the 1 instead?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the dropped screen

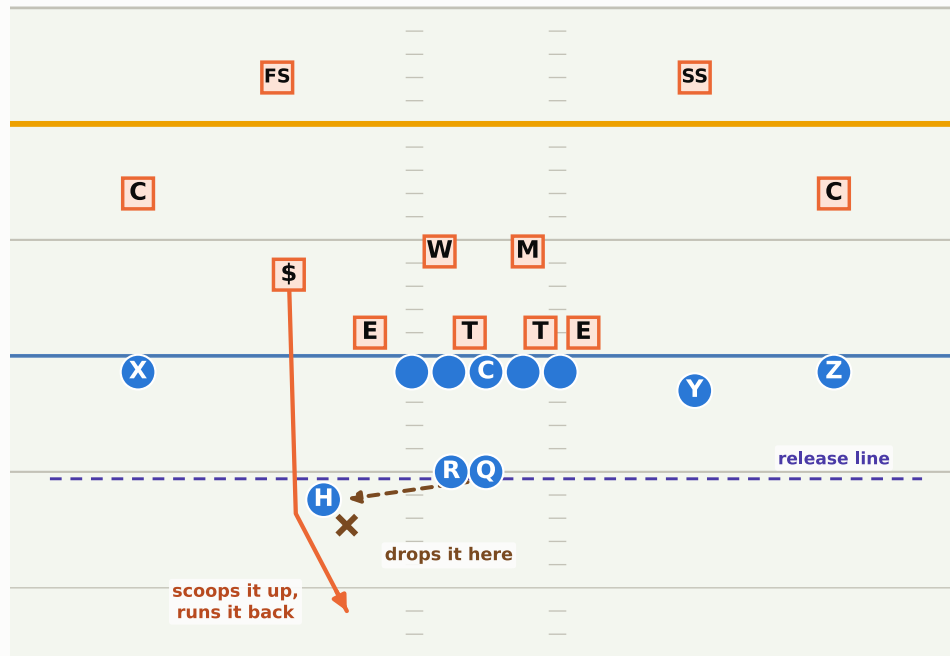


Figure 15: First-and-10. The quarterback, in the shotgun, catches the snap five yards deep and immediately throws to the slot receiver (H), who has bowed back to receive a bubble screen. The receiver is a yard behind the spot where the quarterback released the ball when the pass bounces off his hands and onto the turf. The nickel defender (\$) scoops it up and runs the other way while most players stop.

Do you keep watching the nickel defender? What's the ruling?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Seven officials split the field and the players.** The referee (white cap) watches the quarterback and announces; the umpire watches the interior line; the down and line judges own the line of scrimmage and the sidelines; the side, field and back judges own the receivers downfield. Who threw a flag tells you what it is likely to be.
- **Replay has three doors.** Coaches challenge (two, a third if either succeeds since 2024, a timeout lost if they fail, never inside two minutes of a half). The booth reviews scores, turnovers, tries and the final two minutes. Replay assist (since 2021) fixes objective facts and, since 2024, wipes out certain flags; since 2026 the league may consult on ejections. Judgment fouls aren't reviewable, and there is no sky judge. Changes require clear and obvious evidence, so a hidden moment usually "stands".
- **A catch is three gates: control, feet down, then time** (an act common to the game, or long enough to make one). The 2018 rewrite removed the survive-the-ground clause after the Johnson, Bryant and James rulings, trading an objective test for one that matches what people see.
- **The order of events decides possession.** Ball out before the knee is a fumble; knee down after contact first is a tackle; untouched, an NFL runner isn't down. Forward progress sets the spot. Officials hold the whistle on close fumbles because replay can undo a fumble but can't invent a recovery.
- **The rules of the ball are about direction and territory.** A forward fumble out of bounds comes back to the spot of the fumble, but into the opponent's end zone and out it's a touchback. A pass that doesn't go forward, parallel included, is a backward pass and stays live if dropped. Grounding is a foul only inside the tackle box, or short of the line; once a passer has been outside the box, a throwaway just has to reach the line.
- **Cameras measure, people spot.** Hawk-Eye has replaced the chain walk since 2025, but it measures from a spot an official chose by eye. Crew differences in flags are real but small and unstable from year to year.

Connections

Builds on: Contact Rules and Penalties (the fouls these officials call, and how they are enforced: spot fouls, accept or decline, half the distance) and Quarterback Play (the throwaway, and why a quarterback leaves the pocket). The spot, forward progress and the chain crew were introduced in Downs, Distance, and Field Position; the end zone and the plane of the goal line in The Game in Fifteen Minutes; the touchback on kicks in Field Goals, Extra Points, and the Punting Game; the challenge decision and the staff who advise on it in What a Play Really Is; and why tempo offenses hurry to beat a challenge in In-Game Adjustments and Tempo.

Sets up: The Clock, the Two-Minute Drill, and Fourth Down, where spikes, grounding, the booth's control of the final two minutes and the value of a timeout spent on a challenge all become clock decisions; and Red Zone, Goal Line, Two-Point Plays, and Short Yardage, where forward progress and the goal-line spot decide the tush push and every fourth-and-inches. The rules' history (the 1978 passing changes, the catch rule, replay) is placed in context in The 1978 Liberation and Appendix E, and Part 13 shows how to pull the review and penalty data behind this chapter's charts yourself (nflverse in Python).

i Go deeper

- The **NFL Rulebook** (NFL Football Operations, free online): Rule 3 (definitions, including the catch and the pocket area), Rule 7 (dead ball, down by contact), Rule 8 (forward and backward passes, intentional grounding, fumbles), Rule 11 (touchbacks) and Rule 15 (instant replay). The league also publishes officiating videos explaining recent rulings, with the same angles the replay officials use.
- **Football Zebras** (footballzebras.com), written by former officials and longtime followers of officiating: crew assignments, rule changes and play-by-play explanations of controversial calls.
- **nflverse** (via `nflreadpy`): `load_officials()` for every game's crew since 2015 and the play-by-play `desc` text for every recorded review and penalty, enough to rebuild both charts in this chapter and test your own theories about crews.
- Former NFL officiating chiefs (Mike Pereira, Dean Blandino) explain controversial calls on broadcasts and in columns every week; following them is the fastest way to learn how the people in the booth think.

Answers to Predict the play

Answer 1.

He can't challenge: it's inside the final two minutes of the half, so only the booth can review it, and the coach's flag would be wasted (a coach who throws one when he isn't allowed to is charged a timeout, or 15 yards if he has none left). The replay official decides whether to stop the game; if he does, New York rules, and the referee announces it.

The review runs the three gates. **Control:** did he have the ball secured before it touched anything? **Feet:** did both feet come down in bounds, the right and then the left, while he had control? If yes, the first two gates are passed. **Time:** a toe-tap along the sideline leaves no time for an act common to the game before he goes out, so he had to **keep control through his landing out of bounds**. The ball came loose when he hit the ground, so if the video shows it, the answer is **"the ruling on the field stands"** or **"is confirmed": incomplete**. If the video instead shows him landing with the ball still secured and it coming loose only as he rolled over afterward, it's **reversed: a catch**, and because he was out of bounds the ball is dead there, not a fumble.

Answer 2.

A fumble through the end zone: touchback. The ball was fumbled forward into the opponent's end zone and went out of bounds there, so the defense takes over at its own 20, even though it never recovered the ball. Because it's a turnover, the booth reviews it automatically; the only questions are whether the ball came out before the runner's knee or body touched the ground (if his knee was down first, he was down by contact at the 1, and it's fourth-and-goal from there) and whether it crossed the goal line before going out.

If the ball had gone out of bounds *at the 1*, short of the goal line, the offense would keep it. And because a fumble that goes out forward comes back to the spot of the fumble, it would be at the 1, where the punch knocked it out: fourth-and-goal. The difference between those two outcomes is a few feet of turf and roughly a touchdown's worth of expected points, which is why coaches tell runners to carry the ball high and tight near the goal line instead of reaching for the pylon.

Answer 3.

Keep watching. The receiver was behind the yard line where the quarterback released the ball, so the pass traveled backward: it was a **backward pass**, not a forward pass. A dropped backward pass is a live ball, like a fumble, so the defender's recovery counts

and he may advance it. If he scores, it's a defensive touchdown (and, as a change of possession and a score, the booth reviews it, looking at exactly one thing: whether the ball traveled forward). The officials will usually let this play run rather than blow a whistle, for the reason in the 2008 story: replay can take a touchdown away cleanly, but can't hand the defense a recovery it was whistled out of.

Had the receiver caught it a yard *ahead* of the release point, which is where most shotgun screens are caught, the same drop would be an incomplete forward pass and a dead ball. Defenders are coached to scoop up every dropped screen for exactly this reason.